Example 2.6

Find the perimeter and area of the given Fig.2.23. $\left(\pi = \frac{22}{7}\right)$

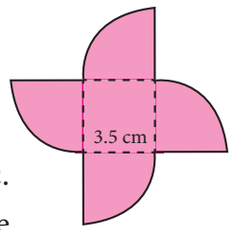


Fig. 2.23

Solution:

Radius of a circular quadrant, $r = 3.5\ cm$ and side of a square, $a = 3.5\ cm$.

The given figure is formed by the joining of 4 quadrants of a circle with each side of a square. The boundary of the given figure consists of 4 arcs and 4 radii.

(i) Perimeter of the given combined shape

$= 4 \times$ length of the arcs of the quadrant of a circle $+ 4 \times$ radius

$$= \left(4 \times \frac{1}{4} \times 2\pi r\right) + 4r$$

$$= \left(4 \times \frac{1}{4} \times 2 \times \frac{22}{7} \times 3.5\right) + (4 \times 3.5)$$

$= 22 + 14 = 36\ cm$ (approximately)

(ii) Area of the given combined shape

= area of the square + 4 × area of the quadrants of the circle

$$= a^2 + \left(4 \times \frac{1}{4}\pi r^2\right)$$

$$= (3.5 \times 3.5) + \left(\frac{22}{7} \times 3.5 \times 3.5\right)$$

$A = 12.25 + 38.5 = 50.75\ cm^2$ (approximately)

Example 2.7

Nishanth has a key-chain which is in the form of an equilateral triangle and a semicircle attached to a square of side 5 *cm* as shown in the Fig. 2.24. Find its area. $\left(\pi = 3.14, \sqrt{3} = 1.732\right)$



Fig. 2.24

Solution:

Side of the square $= 5\ cm$

Diameter of the semi-circle $= 5\ cm \Rightarrow$ Radius $= 2.5\ cm$

Side of the equilateral triangle $= 5\ cm$

$\therefore$ Area of the key-chain = area of the semi circle + area of the square + area of the equilateral triangle

$$= \frac{1}{2}\pi r^2 + a^2 + \frac{\sqrt{3}}{4}a^2$$

$$= \left(\frac{1}{2} \times 3.14 \times 2.5 \times 2.5\right) + (5 \times 5) + \left(\frac{\sqrt{3}}{4} \times 5 \times 5\right)$$

$$= 9.81 + 25 + 10.83$$

$= 45.64\ cm^2$ (approximately)

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